

LearnWave — Building an AI-Native Learning Platform

How I used Generative AI to design, build, and ship a production LMS

Next.js 15 · TypeScript · Prisma + NextAuth · Google Gemini AI · Web Speech API

3

user roles served
(Student / Mentor / Admin)

2

AI-powered features shipped
(academic assistant + voice nav)

15+

technologies integrated
(Next.js, Prisma, Gemini, etc.)

The Problem

As team lead for LearnWave, a production-ready Learning Management Platform, I set out to fix what generic LMS tools get wrong: static content, no adaptive support, and clunky navigation across very different user types — students, mentors, and admins. I scoped AI as one of three core product pillars from day one, alongside role-based dashboards and real-time collaboration — not as a bolt-on chatbot.

How I Leveraged AI

Product Design	Integrated Google Gemini AI as an in-app academic assistant — giving students subject-specific help, auto-generated study plans, and on-demand concept explanations, built as a core dashboard feature rather than an afterthought chatbot.
Accessibility & UX	Designed and shipped voice-based navigation on the Web Speech API, letting students, mentors, and admins move through the entire platform hands-free using role-specific commands — a deliberate call to reduce friction for a non-technical user base.
Speed to Prototype	Used AI coding assistants throughout the build to go from concept to a working, role-based platform — three dashboards, session booking, live analytics — far faster than a traditional build cycle would allow a student-led team.

Outcome & Takeaway

The result: a working, role-based platform with an integrated AI assistant, hands-free voice navigation, live analytics dashboards, and a mentor-booking system — not a static course-listing app. It gave a student-led team a genuinely differentiated product at a fraction of a traditional engineering timeline.

My biggest takeaway as a fresher: AI shouldn't just accelerate execution — it should shape the product decisions themselves. I ask where AI creates genuine user value first, then use AI tools to get there faster.